

Referential Integrity (or foreign key) Constraint

- Statement of the constraint
 - The value in the foreign key column (or columns) FK of the the **referencing relation** R1 can be **either**:
 - (1) a value of an existing primary key value of a corresponding primary key PK in the **referenced relation** R2, or
 - (2) a **null**.
- In case (2), the FK in R1 should **not** be a part of its own primary key.

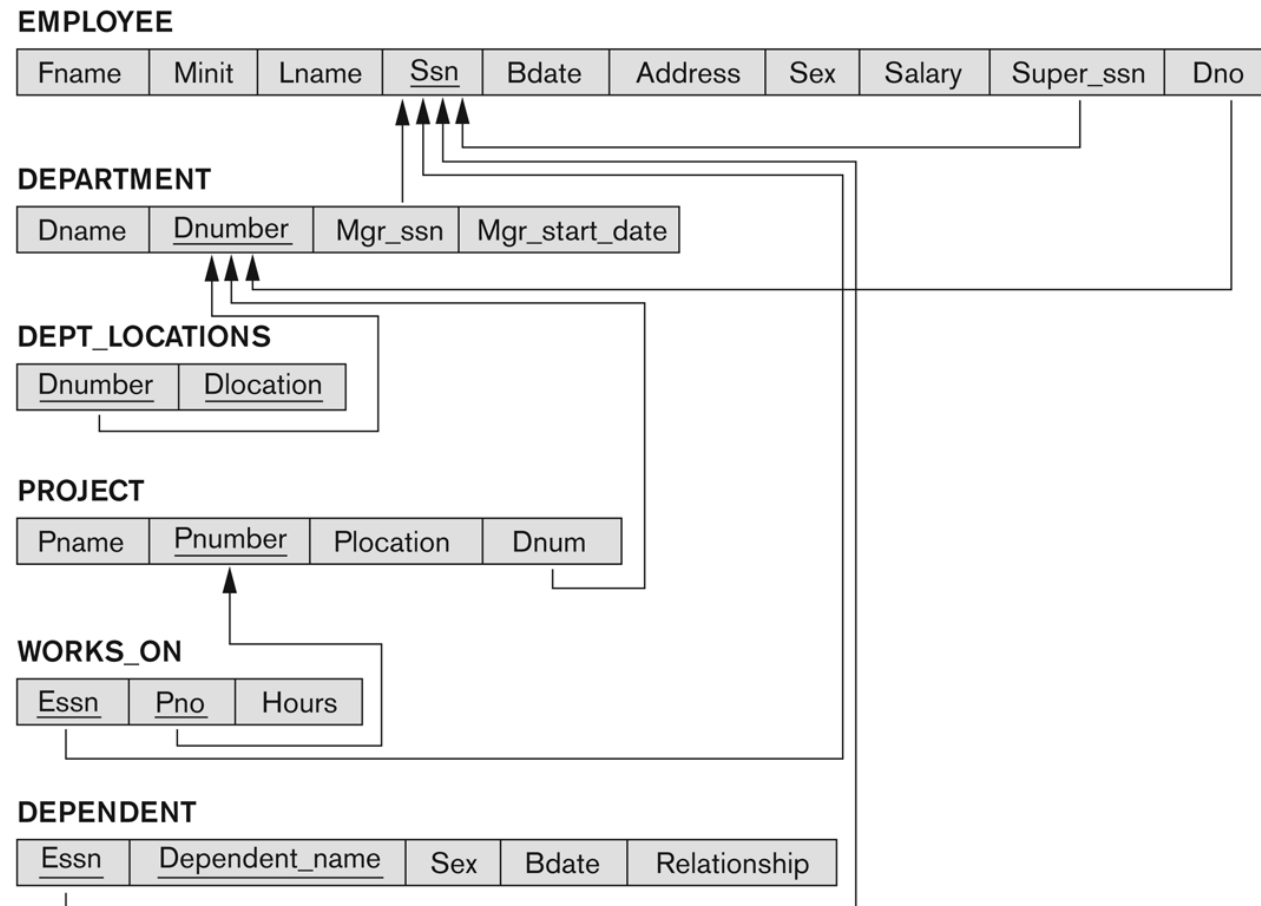
Displaying a relational database schema and its constraints

- Each relation schema can be displayed as a row of attribute names
- The name of the relation is written above the attribute names
- The primary key attribute (or attributes) will be underlined
- A foreign key (referential integrity) constraints is displayed as a directed arc (arrow) from the foreign key attributes to the referenced table
 - Can also point the the primary key of the referenced relation for clarity
- Next slide shows the COMPANY **relational schema diagram**

Referential Integrity Constraints for COMPANY database

Figure 5.7

Referential integrity constraints displayed on the COMPANY relational database schema.



Displaying a relational database schema and its constraints

- Each relation schema can be displayed as a row of attribute names
- The name of the relation is written above the attribute names
- The primary key attribute (or attributes) will be underlined
- A foreign key (referential integrity) constraints is displayed as a directed arc (arrow) from the foreign key attributes to the referenced table
 - Can also point the the primary key of the referenced relation for clarity
- Next slide shows the **COMPANY relational schema diagram with referential integrity constraints**

Other Types of Constraints

- **Semantic Integrity Constraints:**
 - based on application semantics and cannot be expressed by the model.
 - Example: “the max. no. of hours per employee for all projects he or she works on is 56 hrs per week”
- A **constraint specification** language may have to be used to express these
- SQL-99 allows **CREATE TRIGGER** and **CREATE ASSERTION** to express some of these semantic constraints
- Keys, Permissibility of Null values, Candidate Keys (Unique in SQL), Foreign Keys, Referential Integrity etc. are expressed by the **CREATE TABLE** statement in SQL.

Update Operations on Relations

- INSERT a tuple.
- DELETE a tuple.
- MODIFY a tuple.
- Integrity constraints should not be violated by the update operations.
- Several update operations may have to be grouped together.
- Updates may **propagate** to cause other updates automatically. This may be necessary to maintain integrity constraints.

Update Operations on Relations

- In case of integrity violation, several actions can be taken:
 - Cancel the operation that causes the violation (RESTRICT or REJECT option)
 - Perform the operation but inform the user of the violation
 - Trigger additional updates so the violation is corrected (CASCADE option, SET NULL option)
 - Execute a user-specified error-correction routine

Possible violations for each operation

- INSERT may violate any of the constraints:
 - Domain constraint:
 - if one of the attribute values provided for the new tuple is not of the specified attribute domain
 - Key constraint:
 - if the value of a key attribute in the new tuple already exists in another tuple in the relation
 - Referential integrity:
 - if a foreign key value in the new tuple references a primary key value that does not exist in the referenced relation
 - Entity integrity:
 - if the primary key value is null in the new tuple

Possible violations for each operation

- DELETE may violate only referential integrity:
 - If the primary key value of the tuple being deleted is referenced from other tuples in the database
 - Can be remedied by several actions: RESTRICT, CASCADE, SET NULL
 - RESTRICT option: reject the deletion
 - CASCADE option: propagate the new primary key value into the foreign keys of the referencing tuples
 - SET NULL option: set the foreign keys of the referencing tuples to NULL
 - One of the above options must be specified during database design for each foreign key constraint

Possible violations for each operation

- UPDATE may violate domain constraint and NOT NULL constraint on an attribute being modified
- Any of the other constraints may also be violated, depending on the attribute being updated:
 - Updating the primary key (PK):
 - Similar to a DELETE followed by an INSERT
 - Need to specify similar options to DELETE
 - Updating a foreign key (FK):
 - May violate referential integrity
 - Updating an ordinary attribute (neither PK nor FK):
 - Can only violate domain constraints

Summary

- Presented Relational Model Concepts
 - Definitions
 - Characteristics of relations
- Discussed Relational Model Constraints and Relational Database Schemas
 - Domain constraints
 - Key constraints
 - Entity integrity
 - Referential integrity
- Described the Relational Update Operations and Dealing with Constraint Violations